

CASE 3

Aspiring Quants

White Stone is an internationally recognized hedge fund, boasting its high return yield to investors and a highly selective hiring process. Every year, White Stone receives 22,000 applications for their analytics division but hires only 300 applicants. Most of these applicants are fresh graduates coming from Ivy League schools, and career shifters who have years of experience.

For years, fundamental analysis has been working for White Stone. With AI coming into the forefront of many business innovations, the age of using these technologies to trading is nearing.

“There must be a more data-driven way of trading, ‘ya know?” said Naruto Cooper, the fund manager.

“Although we hire a lot of fresh grads, we are ultimately stunted because the ones monitoring them are averse to changing practices that just work. This, in turn kill new ideas.” reacted Killua Uchiha.

“I have a fun idea. Because it’s clear that hiring individuals often get stunted by our more tenured analysts, why not hire teams?” said Bujin Ma, one of the senior analysts of the fund.

“What do you have in mind?” Naruto replied.

“Why not host a hackathon with the intent of hiring the team that can make, not just most accurate model, but also the most profitable one?” said Bujin Ma.

“I like the idea. Maybe we can give them a typical analysis we do. We then hire the team who can give the most creative solution” contributed Killua.

A group of friends

Zelda just graduated from her bachelor's degree in data science. She's been exposed to a couple of projects, done research, and even have experience with several work engagements through her work immersions. Now armed with her degree, she can't help but feel lost.

She kept reminiscing of the times she worked with her friends. She participated hackathons, organized data science events, and even used her knowledge in time series to trade stock and Bitcoin.

Weeks after her graduation, she was lying down on her sofa eating chips while doomscrolling, until she came upon an ad from White Stone. The ad said "Want-a-Quant: Trading Hackathon. Winning team gets to be hired to our company". A sudden rush of adrenaline came over her, and she jumped off from the sofa.

"Is this real? Is this real?" she kept repeating. Immediately, she messaged her friends for a sudden Discord call. Everyone was confused at first, but they quickly understood the moment Zelda shared the ad to their group chat. After a second or two, Discord started ringing.

"Registration ends next week! I can come up with a plan!" exclaimed Armin.

One by one, her friends joined in. "I just want to be free from this house" expressed Eren.

"Let's register now! I want us to be the very best!" said Ash.

"And they just need a team of five!" responded Voltes.

As the friends stayed on the call, they all fixed their resume's, brushed up on their coding, registered, and talked about their college years.

Hackathon day

Weeks of preparation has finally lead to this. Zelda and her team, called the Phantom Troupe, have done their homeworks. They spend day and night studying forecasting and quant mathematics. The hackathon will last for a week. Within this week, White Stone will be providing each team a room, food, and some compensation for all the participants.

The goal of the hackathon is for teams to build a model or more and create a stock portfolio, containing at least 2 stocks, that can accurately predict stock closing prices. However, the organizers do recognize that an accurate model doesn't necessarily mean a profitable one. So there's a separate criteria for the most profitable model.

The winner is one who can balance both accuracy and profitability with their proposed model, portfolio, and strategy. By default, backtesting is going to be the main assessment method for a model's profitability.

White Stone has provided a list of stocks that participants can choose from to build their portfolio. However, groups can add more stocks here as they deem fit.

Stock	Category
AAPL	Technology
MSFT	Technology
NVDA	Technology
AMZN	Technology
GOOGL	Technology
META	Technology
TSLA	Technology
JPM	Finance
BAC	Finance
WFC	Finance
C	Finance
GS	Finance
MS	Finance
KO	Consumer Goods
PG	Consumer Goods
PEP	Consumer Goods
WMT	Consumer Goods
COST	Consumer Goods
CL	Consumer Goods
XOM	Energy

CVX	Energy
COP	Energy
SLB	Energy
EOG	Energy
MPC	Energy
SPY	ETF
QQQ	ETF
DIA	ETF
IWM	ETF
VTI	ETF

Every group will be given a starting capital of \$10,000. There also a couple of assumptions that are provided, such as the stock transaction cost is 0.1% and the risk free rate is 2%.

Another constraint that the organizers imposed is to only use yahoo finance to get the data. Because of the COVID-19 pandemic, the organizers constrained the starting date from January 1, 2022 until January 1, 2025. The organizers would use a random 60-day period after January 1, 2025.

The forecasting task is then to be able to do a 60-day forecast, which is accurate and profitable enough. Although these are the main metrics of success, a deep understanding of their strategy and method would tower everything else.

Starting line

“Because there’s a ton of you, we’ll need you to create a technical report. I and the other analysts will be the ones judging your report” said Naruto.

“To help you out, we’ve created some metrics and strategies for you to work with. We’re expecting that you can understand and interpret the metrics we’ve provided for you. After all, we only get the very best.” said Killua.

As you checked the code, the default strategy is a standard forecast-thresholding strategy where you go all in or all out. They also assume that you put equal weight on all of your stocks.

“You have the rest of the week for the hackathon. We hope you learn and have fun.” said Nagato an analyst keen on giving high-fives.